

Assignment 4

1. Define Greedy Algorithms. Explain its characteristics.

2.

Question 4.1

Find the optimal solution for the fractional knapsack problem where knapsack capacity = 28, $P = \{9, 5, 2, 7, 6, 16, 3\}$ and $W = \{2, 5, 6, 11, 1, 9, 1\}$.

Answer: $S = \{I_5, I_1, I_7, I_6, I_2, 0.91 \times I_4\}$, $SW = 28$, $SP = 45.37$.

3.

Question 4.2

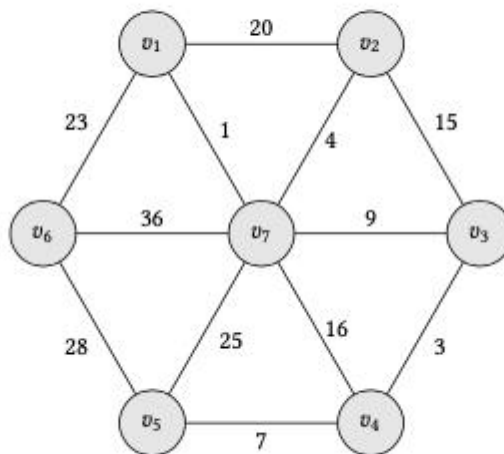
Solve: $n = 7$, Profits $(p_1, \dots, p_7) = (3, 5, 20, 18, 1, 6, 30)$, Deadlines $(d_1, \dots, d_7) = (1, 3, 4, 3, 2, 1, 2)$.

Answer: $\{J_7, J_3, J_4, J_6\}$, Profit = 74 units.

4.

Question 4.3

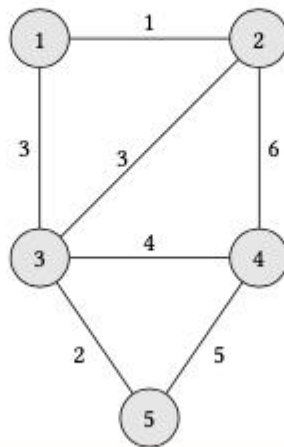
Find out the minimum cost-spanning tree using the KRUSKAL algorithm.



5.

Question 4.4

Find the cost of Minimal Spanning Tree of the given graph by using Prim's Algorithm.



6.

Assignment:

Suppose Dijkstra's algorithm is run on the following graph, starting at node A. Find the final shortest path tree.

